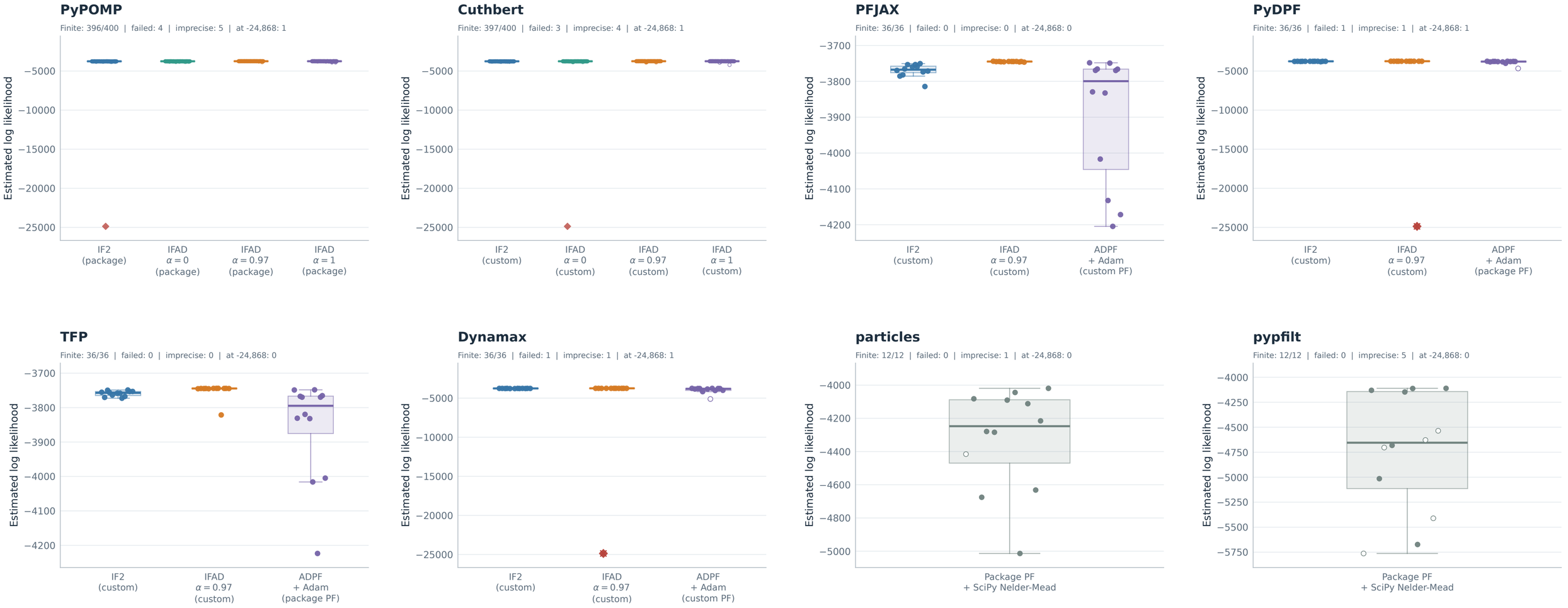
Dacca: log likelihood estimates

Full range, with separate vertical scales for each package. Counts include failed optimizations.

PyPOMP and Cuthbert: 100 initial values per method. Other packages: 6 initial values x 2 replicates. PyPOMP PF likelihood evaluation, J=5,000.



Package denotes code supplied by the library; custom denotes code added for this study. Dynamax uses package SSM.fit_sgd with a custom PF. Package PF alone is not a package parameter-estimation method.

ADPF + Adam: differentiable PF likelihood, alpha 1, 425 updates without IF2 initialization. IFAD starts with IF2; it is a different optimization procedure.

Boxplots include all finite log likelihood estimates. Bars show +/- 1.96 MCSE. Failed optimizations are plotted at the last valid parameters where available.

Densities below 10^{-18} are replaced by 10^{-18}. Log likelihood -24,868 corresponds to this bound at all 600 observations; MCSE=0 at this value results from the numerical bound.